AERIS Expert Labeling Guide

DRAFT Version 3 (changes from version 2: the evidence table now prints the 6-hourly means and per-station means the models were given, and the station map has distance rings, so claims citing those numbers are checkable; the compound-claim rule is narrowed; it now says plainly that a language model wrote the claims. No label definition, threshold, or recording step changed.) This guide explains how to fill out a blinded AERIS claim packet. The example packet that comes with it is just for checking the format; nothing counts as an official label until my freeze checks pass.

What your labels are for

Every claim in this packet was written by a large language model. Each model was shown one anomaly and the stored measurements around it, and asked to reason through the physical cause in four steps. The claims are the sentences that came out. They are what is being tested here — not something AERIS believes.

AERIS then runs two automatic checks on every model-written sentence. The first asks whether the sentence is actually grounded in the data the model was shown. The second scores each claim against independent measurement channels using fixed physical rules, with no language model and no human involved. That second score is designed to work without an expert in the loop, but before I can claim it works, I have to test it against real expert judgment once. Your marked packet is that test: your labels become the testing dataset, and the result I report is how well the automatic score agrees with you. The scoring pipeline never sees your labels, and none of its rules or thresholds are tuned on them, so the comparison stays honest. This is a one-time validation, not a review step that stays in the product. The final product runs on its own: it detects an unusual reading, has a model explain it, checks the explanation automatically, and shows the user the explanation with its verification result.

What's in the packet

Each case has four parts, in this order:

1. **Anomaly.** The basics of the event: data source and metric, observed and expected values, UTC time, location, severity, and which detectors flagged it.
2. **Stored evidence summary.** A table of the sensor data stored for the 72 hours around the event, then a station map and wind panels drawn from that same data. Each source and metric gets one row with its range, mean, and the reading nearest the event. Indented underneath it are two more rows: the metric's **6-hourly means** across the window, and its **per-station means** with each station's distance from the event. Those two rows are there because the models were given exactly them — so if a claim cites a 6-hour average or a distance, you can check the number instead of having to take it on faith.
3. **Claims.** The statements you're labeling, each with its own row of V / I / U boxes.
4. **Optional: most likely cause.** A blank box at the end for your own best guess at what caused the event. You can skip it.

If different models wrote the same claim word for word, it's printed once. The claims are shuffled, and the shuffle is different for each labeler, so your packet and mine are in different orders. Each claim has a number printed next to it, and that number is how I'll refer to the claim if I ever need to ask you about one.

You'll never see which model wrote a claim, how sure that model was, or any score from my pipeline. That's on purpose: nothing in the packet should nudge you toward or away from what the software thinks.

The data sources, briefly

These names show up in the evidence table and on the plots:

- OpenAQ: a worldwide aggregator of government air quality monitors. I only include stations I could verify as real regulatory monitors.
- TCEQ: Texas state monitoring stations (NO_2, SO_2, CO), read from the commission's public site.
- PurpleAir: cheap optical PM2.5 sensors that people mount on their houses. Dense coverage, noisier data; they go through their own quality screen.
- ASOS: the automated weather stations at airports. Direct anemometer and thermometer readings.
- OpenWeather: a commercial weather product that blends several inputs, including station data.
- NOAA GFS: the global forecast model. I use its f000 analysis fields (winds, boundary layer height). It assimilates surface observations, so it isn't independent of ASOS.
- Sentinel-5P: the ESA satellite that measures column NO_2, SO_2, and other gases from orbit.
- EPA AQS: EPA's official archive of ground chemistry (NO_2, SO_2, CO). Only used for the historical window, not live collection.

Label definitions

Mark each claim you answer with exactly one of these:

- **Valid (V):** the claim holds up at the precision it states, and the evidence in the packet supports it.
- **Invalid (I):** the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning doesn't hold.
- **Unsure (U):** you can't tell from what's shown. This covers thin evidence, measurements that can't resolve the claim, ambiguous wording, and anything outside what you're comfortable judging.

Two rules that follow from these:

- Missing or thin evidence means Unsure, not Invalid.
- A cause that's plausible but unproven stays Unsure unless the evidence actually contradicts it.

How to mark claims

You can annotate the PDF digitally, or print it, mark it by hand, and scan it back. Either works; both count the same.

1. Mark exactly one box (V, I, or U) for each claim you answer.
2. To skip a claim, leave all three of its boxes blank. I record a blank as missing; I never quietly turn it into Unsure.

3. For Invalid or Unsure, add a short note when the reason isn't obvious. Notes on Valid are welcome but optional.
4. Some claims bundle several statements into one sentence. Judge them by their parts. If every part earns the same verdict, use that verdict — an all-correct compound claim is Valid, and a compound claim whose parts are all contradicted is Invalid. Mark Unsure when the parts would earn **different** verdicts, when the wording is ambiguous, or when the claim leans on something it never actually states. Add a few words saying which part drove your call. Every claim stays in the study, even the messy ones.
5. The "most likely cause" box at the end is optional, and "insufficient evidence to determine a single cause" is a perfectly fine answer.

Reading the plots

The station map: the black star is the anomaly, and the colored dots are every station that contributed data in the stored 72-hour window. The legend gives the station count per source. The dashed rings are great-circle distance from the event, labeled in km at the top of each ring, so you can read a station's distance straight off the map; a ring that would fall outside the plotted area is left off. The rings look slightly oval because the axes are longitude and latitude, and a degree of longitude is shorter than a degree of latitude at this latitude — the rings are true constant-distance curves regardless. There's no basemap on purpose: nothing on this map is context the models didn't have.

The wind panels: one panel per wind source (ASOS, GFS, OpenWeather). Each arrow is one station's wind reading closest to the event time, drawn pointing downwind. Longer arrows mean faster wind, but each panel has its own scale, so only compare arrow lengths within a panel, never across panels. The label above each panel gives the speed of its longest arrow. ASOS and OpenWeather arrows come from stored speed and direction values; GFS arrows come from its stored u and v wind components.

Everything in the evidence table and on both plots comes from the same stored data the models saw. An automated audit re-derives all of it from the database and refuses to release a packet that has drifted, including one that has dropped any of the 6-hourly or per-station rows.

Calm-wind flags

Some wind-direction claims have an orange flag printed under them. The flag means at least one relevant wind source reported an event-time speed below that source's calm cutoff, so there may not have been a stable wind direction to reason from. (Transport claims check GFS, OpenWeather, and ASOS; point-source claims check GFS and OpenWeather.) The flag shows each affected source's event-time speed, the cutoff it used, and how many wind readings that cutoff came from, so you never have to compute anything. The flag is information, not a verdict: a flagged claim is still yours to judge, and you can mark it V, I, or U like any other claim.

The cutoff is computed separately for each wind source: take that source's wind speeds over the stored 72-hour window, compute mean - 2*SD (population SD), and if the result comes out below 1.5 m/s, use 1.5 m/s instead. So the cutoff is max(mean - 2*SD, 1.5 m/s), and a claim is flagged only when the event-time speed is strictly below it. You confirmed the 1.5 m/s floor on July 24, so this rule is final.

Missing or stale wind data never creates a calm flag. It just falls under the normal thin-evidence-means-Unsure rule.

Returning the packet

The marked-up PDF you send back is the official record, and I keep the file exactly as you return it. I type your marks into the database twice, on two separate days, without looking at the first pass while doing the second, then compare the two runs and settle any mismatch against your PDF. If one of your marks is ambiguous, I won't guess what you meant; I'll email you a short numbered list of the unclear ones and wait for your answer. Claims you skipped stay recorded as missing, exactly as marked.

Versioning

Once official labeling starts, this guide is frozen. If any definition, threshold, or step in how I record your marks has to change after that, everything stops: I bump the guide's version number, tell you exactly what changed, and rerun the earlier labels through the pre-declared sensitivity check to show whether the change would have flipped any of them.

When labeling starts

Every rule in this guide is now settled. Official labeling starts once my remaining freeze checks pass and I send you the official packet; until then, nothing you mark on the example packet counts as a label.